

COSMOLOGY ACROSS CULTURES

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These are the proceedings of “Cosmology across Cultures: An International Conference on the Impact of the Study of the Universe in Human Thinking” organized by the Spanish Institutes of Astrophysics of the Canary Islands and Andalucía under the patronage of the European Society for Astronomy in Culture (SEAC). The conference hosted in the multicultural historical city of Granada more than 80 participants from all the continents.

This conference joined specialists of cultural astronomy studies and modern cosmology in a single forum where ideas about the comprehension of the Universe across time, space, and cultures were interchanged, analyzed, revised, and challenged. An experiment, excellently represented by this book, it worked out in a most satisfactory ambience, permitting both modern cosmologists to receive an insight of how people in the past perceived the cosmos and cultural astronomers to understand the great advances of cosmology in the last few decades and consequently the exact and modest position occupied by humankind in an expanding universe dominated by dark matter and dark energy.

The volume is chronologically organized, beginning with modern cosmological studies, followed by historical documentation, and ending with information from archaeological remains. Each section of the book can be studied independently, although a general inspection of the complete volume is recommended to get a correct insight of the spirit of the conference.

This book is of interest to any scholar or student wishing to understand the evolution of the human comprehension of the universe.



VOLUME
409

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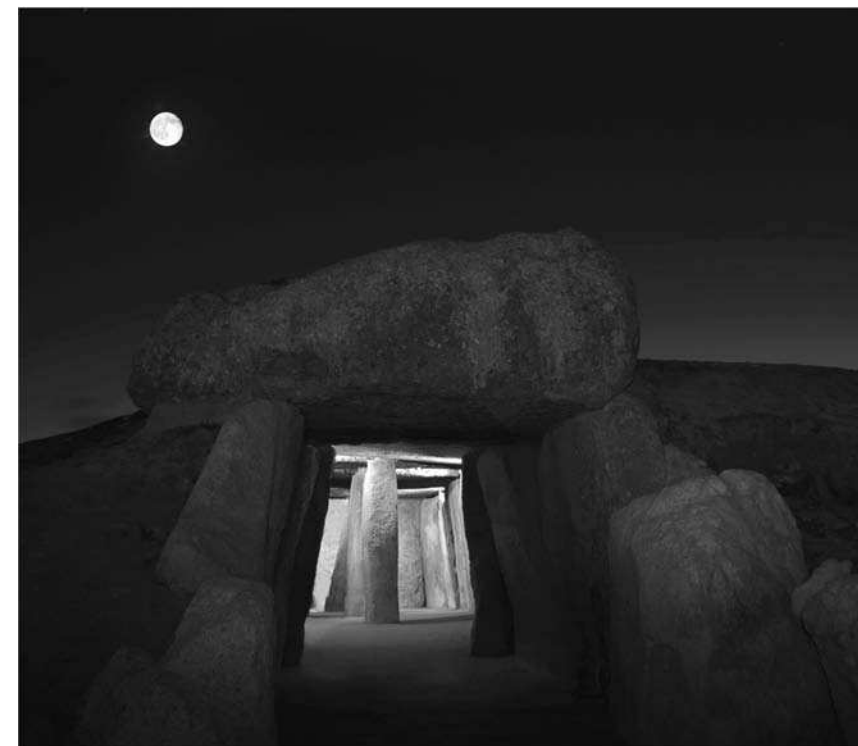
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The Primeval Zodiac: Its Social, Religious, and Mythological Background

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Abstract. In this brief paper we try to draw the lines of the possible development of the originary iconographic and symbolic repertoire of the Mesopotamian zodiac, which through the Greeks was adopted in the Western world.

1. General Introduction

The sky is a wide stage, whose curtain raises for those who look upwards. Most of its protagonists and events, that take place there, are visible all over the globe; others, just in large parts of it; some, only in specific areas. However, being this spectacle silent, without explanations or subtitles, its interpretation is left to one's own imagination. When observing the same celestial phenomena, their perception might be similar in different cultures, their interpretation, however, is deeply related to factors of religious, cultural, economic, and social nature.

Though the idea of the "zodiac" and the zodiacal belt developed independently in different cultures, the one known and used nowadays, which was adopted by the Western cultures through the Greeks, is of Mesopotamian origin. Its use can be found in the Second and First Millennium Babylonian tradition van der Waerden (1952/53); Brack-Bernsen & Hunger (1999), but its origin can be tracked back to the Third Millennium, deeply rooted in the Sumerian civilization. Being the Mesopotamian cultures mainly aniconic, - in the sense of a preference for written description instead of graphic representation, most of the Mesopotamian data are indubitably less appealing and immediately understandable as those of other cultures, mainly the Egyptian ones.

In fact, while the names of the constellations are fully documented since the Old Babylonian period at the beginning of the 2nd Millennium BC, representations are few, scattered, and, mainly, concentrated in specific periods. The sudden spread of such representations have encouraged scholars to interpret the religion of these periods as "astrolatry". This is the well known case of the *kudurrus* (fig. 1), boundary stones recording land grants in the Kassite period (1600-1400 BC), which are covered with astral and divine symbols Seidl (1989); Slanski (2003); the other case is constituted by the materials from the Southern Mesopotamia Seleucid period (312-60 BC), seals (fig. 2) bearing one or more zodiacal signs related to the owner of the seal Wallenfels (1993) it is worth noting that this is the same period and area where the horoscopes in cuneiform writing come from Rochberg (2004), and few drawings (figg. 2, 3, 4) accompanying the description of constellations in astronomical texts written in cuneiform writing on clay tablets Weidner (1967).

2. The Zodiacal Figures in Ancient Mesopotamia

When the sources became homogeneous in form, content and geographical distribution, the zodiac had already been arranged in the shape of the one that modern culture has inherited from the Greeks. But, far from being homogeneous, this zodiac appears to be the canonization of different stratified traditions. A great part of these traditions clearly link up to the agricultural panorama described in the earlier mythical cycle, as well as to the socio-economic world of the earlier Sumerian community.

The myth of the “dying god” Dumuzi (Semitic Tammuz), sought and mourned by his woman Inanna or his sister Geštinanna, seems to be in the root of different figures of the zodiac, as for example the constellations of *Virgo* (fig. 7) and, probably, *Aries*. Several animals involved in the story are represented in the zodiac, but their presence could be justified due to the fact that they also belong to the common agricultural Sumerian background, as the ox (*Taurus*) for its main role in the Sumerian economy, but also related to the “Bull of Heaven” of the myths Sołtysiak (2002), or the Lion Hartner (1965) and the Scorpion for their symbolic and religious values.

Pabilsag (*Sagittarius*), a very important god of the older period belonging to a local pantheon, which had disappeared long time ago from the official worshipping, continues to be remembered as a composite being that maintains the symbols of the war god (fig. 5, 6). Other specific figures are part of the everyday life, as the netherworld duo (*Gemini*), Meslamta’ea and Lugalgirra, demons popularly identified with two unwelcome tax collectors.

These figures have a long tradition and evolution that in some cases can be traced not only in the texts, but also in the iconography. A particularly interesting group is the one related to the “aquatic world”. Some of the figures are linked to the god Enki and its cultic center Eridu, in the south of Mesopotamia. The Sumerian god Enki, later identified with the Akkadian Ea, is the god of the sweet waters which run under the soil and rise from the sources, but he is also the god of wisdom and magic, particularly related with the movements of the stars in the Sumerian myths Verderame (2003).

A cycle of these myths, the oldest known from the Sumerian literature, are centered on him as a creator, but the god maintained an important role also in the successive Babylonian tradition. In fact in the Atramhasis epic he is the one who saves humanity by warning the hero of the deluge; at the beginning of the epic, Ea lives in his temple/residence, the Abzu, a sort of deep water abyss, spending most of his time sleeping. This description fits perfectly with one of the representations of the god as a lying giant in the cylinder seals Porada (1987). The typical representation, however, is that of a seated or standing god, with an *aryballos* between the hands, from which two streams of water flow (fig. 8, 9). Both the models have found their way in the astrological tradition: in fact, Enki/Ea is identified with the constellation of the “Giant” (Sum. *gu.1a*),¹ and his relation with the water survived in the idea of a human figure pouring water, which was adopted by the later traditions as the *Aquarius*.

¹ “The Giant (is) the Lord of the Springs, Ea”, BM 82923: 28 (Walker & Hunger 1977, 30); for other passages see Verderame (2002, 52 fn. 123).

Also one of the symbol of the god, the goat fish (Sum. *suhurmaš*, fig. 10), has been considered worthy of a position in the sky, more over in the zodiac (*Capricorn*). This creature, though as a composite being sums up two aspects of the god Enki, the wisdom and magic (the goat) and the aquatic element (the fish), actually it is identified with a real, although mythologized, animal, the giant carp of the Tigris, already described among the Enki's attendants in the Sumerian literary compositions. This fish, which can normally reach up to 1 meter in length, nowadays is among the Iraqi national dishes, the *masgouf* (figg. 11, 12). If walking along the Tigris bank, in Baghdad, an event which sadly you will hardly experience in the next future, you stop in one of the small restaurants in the promenade, the owner will take you to a pool where gigantic carps swim. Once you have chosen your victim, the owner takes the fish and splits it in two halves with an hatchet.²

This is the sort allotted to Tiamat, one of the ancient gods, identified with the salty waters and represented as a marine monster. In the *Enūma eliš* the victorious Marduk splits her body like a *masgouf* or like a fish to be dried (fig. xxx), and proceeds to the creation. The two halves of the body tied together will constitute the sky and the earth, while from her two eyes Marduk makes flow the rivers Tigris and Euphrates. Originally the two rivers, main source of water and life for the region, were identified with the constellation of the *Cancer*. This is clear from the identification of the Euphrates and the Tigris with different parts of the *Cancer* in the astronomical and astrological texts (Verderame 2002, 52-53):

The stars which stand in front of the Cancer (are) the Tigris, the stars which stand behind the Cancer (are) the Euphrates, the Cancer (is) the Euphrates

This association remained also in the late Babylonian tradition: the *Cancer* is represented as two streams or waters, Sum. *a.meš* (Foxvog 1993, 107) which describe the symbols of the zodiacal constellations. When and how the two streams were substituted or eclipsed by the symbol of the crayfish/crab (Sum. *al.lul, nagar*) at the moment it is hard to say: the relation of a freshwater animal with the Tigris and Euphrates is clear, although it is not less important its independent symbolical meanings, as can be seen in many other cultures (Japan, Moche culture).

All these original "figures" were probably related with the earlier cultic calendar, as their popular background seems to suggest. The astronomical text MUL.APIN of the II mill. lists for each period of the year a constellation for reference, as well as relative theological correspondences Hunger & Pingree (1999). However, this source proves how the theological, mythological thought, and, in connection with them, the constellations and the zodiac, developed during the three Millennia of the Mesopotamian civilization(s): many of the original figures

²For the curious and the gourmet, the method of preparation of the *masgouf* is far from being a complicated nouvelle cuisine recipe: through two wooden peg fixed in the soil the split fish is vertically exposed, not directly, to the fire, and left near the fire for more than one hour, during which you can enjoy a walk along the Tigris, foretasting the dinner or pondering over the fate of the fish.

were re-elaborated and adapted to new realities, others were not yet understood and, as a consequence, “recreated” or misunderstood, as it seems to have been the case of the sign of the Aries, called the “Hired Man”, but identified with a ram from the First Millennium on.

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Figure 1. Kudurru

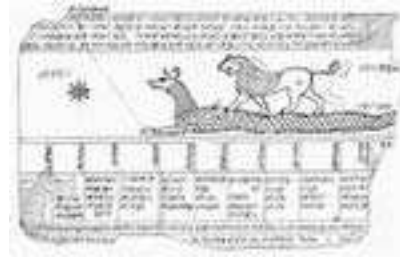


Figure 2. Seleucid tablet

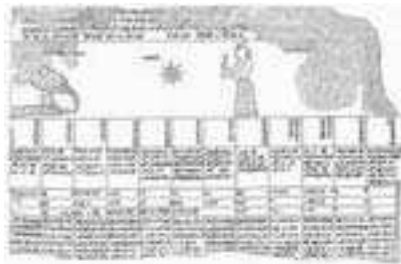


Figure 3. Seleucid tablet



Figure 4. Seleucid tablet

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Figure 5. Seleucid seal



Figure 6. Kudurru



Figure 7. Seleucid seal



Figure 8. Seleucid seal



Figure 9. Seal



Figure 10. Seleucid seal



Figure 11. Masgouf



Figure 12. Uruk IV pictogram of a dried fish